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How Self-Regulated Learning Is Affected by Feedback Based on Large Language Models: Data-Driven Sustainable Development in Computer Programming Learning

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Abstract: Self-regulated learning (SRL) is a sustainable development skill that involves learners actively monitoring and adjusting their learning processes, which is essential for lifelong learning. Learning feedback plays a crucial role in SRL by aiding in self-observation and self-judgment. In this context, large language models (LLMs), with their ability to use human language and continuously interact with learners, not only provide personalized feedback but also offer a data-driven approach to sustainable development in education. By leveraging real-time data, LLMs have the potential to deliver more effective and interactive feedback that enhances both individual learning experiences and scalable, long-term educational strategies. Therefore, this study utilized a quasi-experimental design to examine the effects of LLM-based feedback on learners' SRL, aiming to explore how this data-driven application could support learners' sustainable development in computer programming learning. The findings indicate that LLM-based feedback significantly improves learners' SRL by providing tailored, interactive support that enhances motivation and metacognitive strategies. Additionally, learners receiving LLM-based feedback demonstrated better academic performance, suggesting that these models can effectively support learners' sustainable development in computer programming learning. However, the study acknowledges limitations, including the short experimental period and the initial unfamiliarity with LLM tools, which may have influenced the results. Future research should focus on refining LLM integration, exploring the impact of different feedback types, and extending the application of these tools to other educational contexts.

Keywords: large language models; learning feedback; self-regulated learning; data-driven sustainable development



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1. Introduction

Self-regulated learning (SRL) refers to the learning process in which learners monitor and adjust their learning. By actively evaluating their abilities, developing objectives, and selecting strategies in SRL, learners can promote their metacognition, motivation, knowledge, and skills, and learn how to learn [1]. SRL is not only conducive and necessary in school education but also an essential ability for sustainable development to support lifelong learning [2]. Learners do not always regulate their learning spontaneously and

often struggle to do so effectively in digital and online environments, so providing learning feedback to facilitate SRL activities can improve SRL and overall learning performance [3,4]. Therefore, learning feedback is very important in SRL. Feedback provides learners with information used for self-observation and self-judgment, which guides learners to self-respond more effectively and engage in learning activities, thus influencing learning [5,6]. Research has shown that constant and meaningful feedback is particularly effective in promoting learners' self-efficacy and learning performance [7].

However, even with the support of information technology and visualization, learning feedback remains unidirectional, static, and descriptive. It lacks effective interaction between learners and other entities, which limits its ability to provide continuous and meaningful feedback essential for the self-regulated learning process [8]. The emergence of large language models (LLMs) has enlightened us to address these issues. By analyzing large-scale data on learning progress, LLMs (e.g., ChatGPT) can continuously interact with learners, use human language to answer questions, and personalize interactions based on context to provide learners with different experiences [9].

Our study explored the application of LLMs in SRL through a quasi-experimental study conducted in computer programming learning. The strategy and technical realization of the LLM-based feedback tool are investigated in three aspects: the implementation of LLM-based feedback in SRL; the effect of LLM-based feedback on SRL; and the effect of LLM-based feedback on academic performance. The findings provide practical guidance in applying LLMs to promote the effectiveness of SRL and broaden learners' sustainable development in computer programming learning.

2. Literature Review

2.1. Self-Regulated Learning and Learners' Sustainable Development

In an era of rapid knowledge growth, mastering the skill of sustainable development and effective learning has always been a central focus in education [10]. SRL, a process that is both suitable for and preferred by learners, prepares them for lifelong learning in their respective fields and enables them to stay in a sustainable state of self-improvement and ongoing updates [11]. As a result, SRL is regarded as a crucial skill set for learners' sustainable development in effective lifelong learning across various contexts [2].

SRL is defined as learners' ability to actively shape and guide their learning process in metacognitive, cognitive, motivational, and behavioral dimensions to achieve specific goals, which involves understanding and controlling learning behaviors and environments, making conscious decisions, and pursuing objectives through goal-directed actions, making self-regulation an integral aspect of learners' sustainable development [12–14].

Theoretical models conceptualize SRL as a cyclical process comprising the preparatory, performance, and appraisal phases [12,15]. Self-regulated learners use cognitive processes (summarizing, rereading, and elaboration) to study a topic, and metacognitive processes (orientation, planning, monitoring, and evaluation) to control and monitor their learning and motivate themselves to engage in an appropriate level of learner effort [16]. In other words, self-regulated learners actively monitor and control their learning; they oversee the effectiveness of cognitive and metacognitive strategies they enact throughout a learning session and make decisions to modify these strategies to achieve their goals for learning [1,17,18]. The SRL process emphasizes learners' autonomy, intrinsic motivation, self-monitoring and control, and self-reflection [6]. As a higher-order learning skill, SRL can enhance learners' higher-order cognitive abilities, boost their self-efficacy, and then support their sustainable development in lifelong learning [19].

The link between SRL and improved learning performance has been well-documented in prior research [20,21]. Studies have shown that SRL accounts significantly for the

differences in learners' academic performance in different domains of learning, such as reading comprehension, math problem solving, music learning, athletic practice, etc. [22,23]. SRL has also been found to be related to performance when learners sustainably apply knowledge and skills to a new situation or problem: metacognitive activities promote a deeper understanding of learning content and monitoring activities are associated with the increased use of deep learning strategies [21,24,25].

2.2. Learning Feedback in SRL

In SRL, learners often struggle to spontaneously regulate their learning and experience difficulties in adequately regulating their learning; therefore, feedback plays a pivotal role [26]. Feedback is information regarding aspects of one's performance or understanding provided by an agent to bridge the gaps between the current understanding or performance and a desired goal, aiming to facilitate learning progress, achieve learning goals, and enhance self-regulated skills [8,27].

SRL is closely tied to three key processes: self-observation, self-evaluation, and self-response; feedback provides learners with information for self-observation and self-evaluation, guiding learners to self-respond more effectively and engage in learning activities, thereby influencing learning [28]. In other words, learners establish learning goals based on self-observation and the task, aiming to gauge the gap between their current status and the standards; during self-evaluation, feedback precisely offers diagnostic information to learners regarding whether they have achieved their goals; by leveraging the information provided by external feedback, coupled with monitoring their self-learning process, learners gain clarity on their current learning status and improve subsequent learning strategies, achieving specific goals [29]. Feedback is fundamental in promoting autonomous learning, enabling learners to engage proactively in self-regulation while integrating prior knowledge and experiences. Ultimately, feedback intervention has the potential to shape and modify learners' commitment, motivation beliefs, and the learning process [30].

Feedback can be differentiated based on its cognitive complexity, referring to a task, a process, one's self-regulation, or one's self [31]. Task-level feedback relates to content, facts, or surface information (How well have the tasks been completed and understood? Is the result of a task correct or incorrect?). Process-level feedback concerns performance strategies (What needs to be done to understand and master the tasks?). Self-regulation feedback involves monitoring strategy use without developing new ones (How to manage and monitor actions?). Self-level feedback is about personal qualities in feedback (often praise). Empirical evidence shows that a predominant portion of feedback pertains to task feedback, with the most widely received and comprehended feedback being guidance on the next steps; conversely, self-level feedback tends to be less efficacious [27].

Giving effective feedback to learners is one of the most important responsibilities in SRL, with the rapid development of online learning and artificial intelligence (AI), automatic tools have been proposed to enhance feedback practice [32–34]. Advanced technologies such as AI-driven conversational chatbots, machine learning, and natural language processing (NLP) offer new opportunities to improve the quality of feedback and encourage deeper learner engagement [35]. Open learner models, one of the significant models for enhancing learners' self-regulated learning outcomes, have been empirically assessed in classroom experiments with the support of artificial intelligence [14]. Adaptive learning systems, which offer personalized feedback, can optimize the learning process and improve outcomes by tailoring feedback to individual learners [36]. Scholars have pointed out that AI can facilitate self-regulation and communication in learning by tracking performance in learning assessment and providing feedback on learning outcomes [37].

Currently, most automatic feedback generation primarily relies on methods such as solution comparisons, dashboards, and NLP. However, despite leveraging educational big data and visualization, these feedback mechanisms are still predominantly unidirectional, static, and descriptive, hampering the establishment of an effective interactive dynamic between learners and other entities, which deficiency impedes the facilitation of continuous and meaningful feedback within the context of self-regulated learning [8]. The advent of LLMs has opened new possibilities for addressing these challenges.

2.3. LLMs, a New Data-Driven Application in Education

Large language models (LLMs) are closely linked to data-driven applications as they are trained on massive datasets, learning language patterns and semantics to perform tasks like text generation and reasoning [9,38]. These models enhance data-driven applications by providing strong language understanding across industries, powering services like natural language processing. LLMs can be fine-tuned with new data, allowing for dynamic adjustment and continuous improvement through feedback, making them more personalized and accurate. Additionally, LLMs support data analysis by extracting insights from unstructured data, aiding in decision-making and driving innovation.

In educational areas, LLMs, such as ChatGPT, can help build feedback loops that promote iterative improvement, ensuring that learners not only acquire knowledge but are also equipped with the skills to solve new and evolving problems in learning, which holds significant potential for promoting SRL [39].

Currently, LLMs are being applied in education across three main areas. The first is assessing LLMs' performance in different learning environments. These studies typically use LLMs to answer various questions in specific domains and investigate the potential performance of LLMs in different learning environments [40–43]. For example, researchers examined the ability of GPT-4 to answer 643 self-assessment neurosurgery exam (SANS) committee-style questions from various neurosurgical subspecialties [44]. The results showed that GPT-4 attempted 91.9% of the SANS questions, with an accuracy rate of 76.6%, which means its performance was significantly better than the average of medical learners, neurosurgery residents, and national SANS users.

The second area is exploring the potential impact of LLMs in different learning environments. These studies typically observe and examine how learners use LLMs in the learning process to explore the potential impact of LLMs [45,46]. For instance, Li et al. (2023) conducted a study guiding learners to use ChatGPT in exercises, examining how learners use ChatGPT to complete tasks and assessing ChatGPT's ability to identify and correct web application vulnerabilities [47].

Lastly, researchers are training LLMs to support learning. With their capabilities to generate personalized guidance and explanations, LLMs have the potential to transform how learners access and interact with educational content. Naismith et al. (2023) investigated a method for training GPT-4, which evaluates discourse coherence in a manner consistent with human expert raters [48]. The results of this study demonstrate the significant potential of GPT-4 in generating discourse coherence ratings closely related to human assessments, with clear justifications. Mizumoto et al. (2023) conducted a study examining the feasibility of using GPT-3.5 for automated essay scoring (AES) [49]. The study employed prompts to instruct GPT-3.5 to score given essays, indicating the potential of GPT as a promising tool for AES, contributing to improved scoring. Bezirhan et al. (2023) introduced an innovative approach to generating reading passages using GPT-3 [50]. The results indicate that passages generated by GPT-3 are highly similar to human-authored passages in terms of coherence, relevance, and readability for the target audience. Chen et al. (2023) explored the development of a complete intelligent tutoring system driven by

state-of-the-art LLMs, including automated course planning and adaptation, personalized teaching, and flexible quiz assessment. Their evaluation revealed mechanisms for adaptive behavior, demonstrating the advantages of memory mechanisms and process design in long-term interactions [51]. Dan et al. (2023) developed EduChat, an education-focused chatbot powered by LLMs. Guided by principles from psychology and educational theory, it enhances educational capabilities, including open-ended questioning, essay evaluation, Socratic teaching, and emotional support, all built upon foundational LLMs [52].

2.4. The Contributions of This Study

In this study, we aimed to leverage the capabilities of LLMs (GPT-3.5 model) to provide automated feedback on programming tasks for learners in computer education. This was achieved by integrating the GPT-3.5 model via an application programming interface (API) into an application designed for the automated evaluation of programming code. Specifically, the GPT model was employed to automatically generate tailored feedback for learners enrolled in a programming course. The study presents the findings of a quasi-experimental study that compared a system using human-developed feedback with the system-added GPT-generated feedback and tries to provide practical guidance in applying LLMs to SRL to promote the effectiveness of self-learning and broaden the exploration of LLMs in higher education.

The implementation strategy and technical realization of this study aimed to answer three research questions:

- (1) How is learning feedback based on LLMs implemented?
- (2) Can learning feedback based on LLMs improve SRL?
- (3) Can learning feedback based on LLMs improve academic performance?

3. Materials and Methods

3.1. Participants

The participants of this study were second-year learners in computer science at a university in northern China. They enrolled in an interpreted and object-oriented programming course, Python, which was one of the required courses in their major. The course was based on the object-oriented programming paradigm, providing learners with fundamental concepts and methods in Python development, which included topics such as data types, lists, tuples, dictionaries, functions, exceptions, classes, objects, inheritance, polymorphism, as well as data storage, network programming, and web scraping. It aimed to equip learners with the ability to apply their knowledge of data structures, classes, data storage, network programming, and other concepts in practice, laying a strong foundation for further studies in programming.

Out of 68 learners enrolled in the course, 63 voluntarily participated in the study. They were made aware that their participation in the research study was completely voluntary and they had the right to withdraw from the study at any time without penalty. All their personally identifiable information would remain confidential and would not be included in any publications or presentations.

3.2. Research Design and Data

All participants had learned the C programming language as basic preparation for learning Python and were used to programming code with the code editors on online platforms. The study was supported by RunCode, an online platform designed for the automated execution and testing of programming code.

We ensured that participants were randomly assigned based on their performance in a prerequisite C programming course, which provided prior programming knowledge that

was relevant to the Python course. Specifically, students' final grades in the C programming course were used as a stratification criterion. The participants were then randomly assigned to either the experimental group, which received feedback based on a large language model (LLM), or the control group, which received traditional feedback.

The stratified randomization process involved first categorizing students into three groups based on their C programming grades: low, medium, and high. The grades were divided as follows:

High: final grade of 85 or higher;

Medium: final grade between 70 and 84;

Low: final grade below 70.

Once the students were stratified into these categories, we randomly assigned participants from each category to the experimental or control group. This method ensured that both groups were comparable in terms of their prior programming knowledge, which allowed us to minimize potential biases due to differences in baseline programming skills. Finally, 63 learners were randomly assigned to either the control group ($N = 31$) or the experimental group ($N = 32$).

Data collection in this study encompassed the initial phase of the semester, spanning a 10-week duration out of the 17-week term in 2024. The basic concepts including data types, lists, tuples, dictionaries, functions, exceptions, class, objects, inheritance, and polymorphism were introduced during the first 10 weeks. Thirteen programming assignments covering these concepts were available on the platform to help learners practice the course content. During this study period, all 63 learners submitted their tasks.

While the study acknowledges the relatively short duration of 10 weeks, we would like to clarify that the course was structured with two class sessions per week, which is equivalent to the content typically covered in a 20-week course in many educational systems. Therefore, although the study duration may appear short, the pace and intensity of the course, along with the number of assignments and frequent class sessions, made this 10-week period comparable to a longer study period in a traditional semester. This arrangement is common in certain technical and engineering programs in China, where intensive learning is balanced with practical application. Therefore, the duration was appropriate and sufficient for evaluating SRL (self-regulated learning) improvements in the context of this study.

In the control group, learners only had access to regular support and feedback from the RunCode platform. If the code learners submitted compiled successfully, the feedback provided comprehensive insights into the executed unit tests, encompassing input values and expected output information; if not, in the event of code compilation failure, the platform automatically displayed the standard feedback without necessitating further actions, which encompassed specific details regarding compiler errors. Also, the online code editor within the platform conveniently identified the lines where errors occurred.

In the experimental group, in addition to the standard feedback offered by the platform, learners received supplemental feedback generated by the GPT model through the application programming interface (API), including error identification, code improvement suggestions, concepts and explanations, and hints, which went beyond the regular feedback from the platform.

3.3. Integration of GPT API

In this study, the GPT model was integrated with the RunCode platform via an API to deliver feedback to learners in the experimental group. This integration involved two main components: the front-end interface embedded in the platform and the back-end database. The front-end interface enabled learners to interact with GPT at any time, receiving LLM-

based feedback. The GPT API provided additional support, particularly for programming issues that the RunCode platform could not address, helping learners identify the causes of errors and guiding them in planning their learning paths to enhance their SRL. Besides the interaction with GPT, a prompt interface was included to assist learners in effectively interacting with the model. The back-end database was set up on a server to collect interaction data between learners and GPT, which was later used to explore the impact of LLM-based feedback on learning outcomes.

Errors and exceptions are common challenges learners face when programming in Python. The GPT API generated different types of feedback depending on whether the submitted code contained syntax errors, runtime errors, logical errors, or exceptions, as well as responded to other learning-related queries from learners.

As mentioned above, feedback can be classified into four types: task-level feedback, process-level feedback, self-regulation feedback, and self-level feedback:

- **Task-level feedback.** The feedback from GPT can help learners identify the locations of syntax errors, the types of runtime errors, the discrepancies of logical errors, or the causes of exceptions, and explain the problems and relative concepts, algorithms, or code segments to help learners better understand and program, which belongs to task-level feedback. While the RunCode platform could identify syntax and runtime errors, GPT provided more accessible and easily understood feedback for logic errors, exceptions, and other issues through its anthropomorphic natural language capabilities.
- **Process-level feedback.** GPT can provide suggestions to resolve problems such as helping learners with missing syntax elements, correcting misspellings, checking if variables are correctly initialized, ensuring input validity, adding exception handling, reviewing conditions, loops, or algorithm correctness, modifying specific parts of the code to rectify problems.
- **Self-regulation prompts.** GPT can offer learners hints and guidance to help them self-regulate during problem-solving while avoiding direct provision of answers. This kind of self-regulation feedback is informative and constructive to direct learners toward the right solutions and enhance their comprehension of programming concepts. Importantly, it refrains from disclosing the exact code solutions.
- **Self-level feedback.** Beyond praise, GPT can provide self-level feedback including objective, friendly, and encouraging evaluations that help learners reflect on their learning progress.

These four levels of feedback correspond to the four dimensions of SRL: cognition, behavior, metacognition, and motivation.

Prompting is an emerging interaction technique to support the effective application of LLMs, in which users can employ concise natural language prompts to tell LLMs what to generate [53]. In this study, the four-level feedback of GPT and the characteristics of Python were the basis of prompt design to support learners' SRL in cognition, behavior, metacognition, and motivation:

- Prompts on task-level feedback focused on guiding learners to ask GPT to identify errors and problems of tasks, and requiring more detailed information to improve their cognition in certain aspects, such as: Is my code feasible? What are the exceptions of the code? Please tell me about ... Please check my code.
- Prompts on process-level feedback asked GPT for more guidance to understand and master the tasks. For example: Where to find resources or support of ...? How to understand or master ...? What should I learn or do to master ...?
- Prompts on self-regulation level feedback aimed to give guidance on strategies to improve learning. For example: What strategy should I use to reflect/manage/guide/monitor my behavior in learning ...?

- Prompts on self-level feedback not only related to praise but also tried to give objective and constructive evaluation from GPT to help learners reflect on their learning. For example: I am a learner who is just learning Python: suppose you are my teacher, please give objective and constructive comments and suggestions on my code.

Before the class started, learners in the experimental group were trained to use GPT API on the platform to obtain feedback (in Chinese).

3.4. Instruments and Statistical Methods

First, to identify the influence of LLMs on different dimensions of SRL, we used the revised classic Motivated Strategies for Learning Questionnaire (MSLQ) instrument (Pintrich et al. 1993) [22] and employed the course assignments.

The MSLQ is a classic, 1–7 Likert scale SRL questionnaire, which provides a general cognitive view of motivation and learning strategy (metacognition) to evaluate SRL of college learners. Based on the design principles of the four types of feedback mentioned above, we screened the items in the MSLQ to fit our research characteristics and then changed some words to suit our research environment, such as replacing phrases like “a class” in the MSLQ instrument with references to “the programming platform” to better align with the characteristics of our study. Finally, six items were selected to evaluate learners’ motivation and eight items to evaluate metacognition (please see Appendix A). Also, the study utilized 13 assignments completed during the first 10 weeks of the course. Each assignment was scored out of 100, with the rounded average score providing insights into the learners’ cognitive and behavioral aspects of SRL.

Second, to assess learners’ academic performance, a programming exam (scored out of 100) was administered in the tenth week. Learners in both the experimental and control groups took the exam without support from any tool, ensuring the assessment accurately reflected the knowledge and skills acquired during the first 10 weeks of the Python course.

Finally, to analyze the effects of LLM feedback on SRL and academic performance, several statistical methods were employed. Before conducting any statistical analysis, we used the Shapiro–Wilk test to assess the normality of the data, which is suitable for small sample sizes. We performed the normality test on the average assignment scores, motivation, SRL strategy, and academic performance for each group.

If the Shapiro–Wilk test results indicated that the data followed a normal distribution ($p > 0.05$), an independent samples *t*-test was then used to identify the differences between the experimental and control groups. Additionally, before performing the independent samples *t*-test, we used Levene’s test to assess the equality of variances between the two groups.

If the Shapiro–Wilk test indicated that the data did not follow a normal distribution ($p < 0.05$), we used the non-parametric Mann–Whitney U test to analyze the differences between the experimental and control groups.

4. Results

4.1. The Influence of LLM-Based Feedback on SRL

In this study, learners were randomly assigned to experimental and control groups, each receiving different interventions, resulting in two independent samples. Group descriptives of average assignment scores, SRL strategy, and motivation are shown in Table 1.

The Shapiro–Wilk test was used for the normality test (Please see details in Table 2). Only the results of average assignments scores in both groups were greater than 0.05; therefore, such data were considered to be following the normal distribution, and the data of motivation and SRL strategy in both groups were not following the normal distribution.

Table 1. Group descriptives of average assignments score, motivation, and SRL strategy.

	Group	N	Mean	SD	SE	Coefficient of Variation
Average asgmt. score	Experimental	32	91.500	3.510	0.621	0.038
	Control	31	85.613	4.499	0.808	0.053
Motivation	Experimental	32	5.563	0.982	0.174	0.176
	Control	31	4.968	1.016	0.182	0.205
SRL strategy	Experimental	32	4.250	0.984	0.174	0.231
	Control	31	3.613	1.054	0.189	0.292

Table 2. Test of normality of average assignments score, motivation, and SRL strategy of two groups.

	Average Asmt. Score		Motivation		SRL Strategy	
	Experimental	Control	Experimental	Control	Experimental	Control
Shapiro–Wilk	0.969	0.978	0.865	0.825	0.912	0.881
<i>p</i> -value of Shapiro–Wilk	0.466	0.769	<0.001	<0.001	0.012	0.003

Therefore, Levene’s test of equality of variances and the independent samples T-test were used for statistical analysis of average assignments score to identify the difference in integration of cognition and behavior between the two groups (please see details in Tables 3 and 4), and the Mann–Whitney U test was used for statistical analysis of MSLQ to identify the difference between two groups in motivation as well as in motivation (please see details in Table 5).

Table 3. Test of equality of variances (Levene’s) of average assignments score of two groups.

	F	df ₁	df ₂	<i>p</i>
Average Asgmt. Score	2.354	1	61	0.130

Table 4. Independent Samples T-Test of Average Assignments Score of Two Groups.

	<i>t</i>	df	<i>p</i>	Cohen’s <i>d</i>	SE Cohen’s <i>d</i>	95% CI for Cohen’s <i>d</i>	
						Lower	Upper
Average Asmt. Score	5.801	61	<0.001	1.462	0.311	0.899	2.015

Table 5. Mann–Whitney U Test of motivation and SRL strategy.

	W	<i>p</i>	Rank-Biserial Correlation	SE Rank-Biserial Correlation	95% CI for Rank-Biserial Correlation	
					Lower	Upper
Motivation	656.000	0.020	0.323	0.146	0.047	0.552
SRL strategy	672.500	0.011	0.356	0.146	0.084	0.578

For the Mann–Whitney test, effect size is given by the rank biserial correlation.

Based on the results of Levene’s test of equality of variances in Table 3 ($p = 0.130$), the data were considered to conform to a normal distribution. Also, the results of the T-test in Table 4 showed that there was a significant difference between the experimental group and the control group in average assignments score ($t = 5.801$, $p < 0.001$), which means the experimental group’s cognition and behavior were better than those of the control group, although it presented a moderately small effect size (Cohen’s $d = 1.462$).

Moreover, based on the results of Mann–Whitney U test in Table 5, the experimental group and the control group had significant differences in motivation ($W = 656$, $p = 0.02$,

rank-biserial correlation = 0.323) and SRL strategy ($W = 672.5$, $p = 0.011$, rank-biserial correlation = 0.356), although the two effect sizes were at a medium level. It is concluded that feedback based on LLMs can promote learners' motivation and SRL strategy to some extent.

4.2. The Influence of LLM-Based Feedback on Academic Performance

Group descriptives of academic performance are shown in Table 6. The Shapiro–Wilk test was used to examine the exam scores indicating learners' academic performance in two groups for normality (please see details in Table 7). Although the result of the control group showed the data were in line with normal distribution ($p = 0.889$), the result of the experimental group ($p < 0.001$) was not greater than 0.05, showing the data of the experimental group were considered not to be in line with normal distribution. Further, the distribution in Figure 1 intuitively presents the characteristics of the academic performance data in the experimental group: two subgroups were stratified, one with lower performance, and the other with higher.

Table 6. Group descriptives of academic performance.

	Group	N	Mean	SD	SE	Coefficient of Variation
Academic performance	Experimental	32	78.696	11.085	1.960	0.140
	Control	31	72.065	6.153	1.105	0.085

Table 7. Test of normality (Shapiro–Wilk) of academic performance of two groups.

	Group	W	p
Academic performance	Experimental	0.864	<0.001
	Control	0.983	0.889

Significant results suggest a deviation from normality.

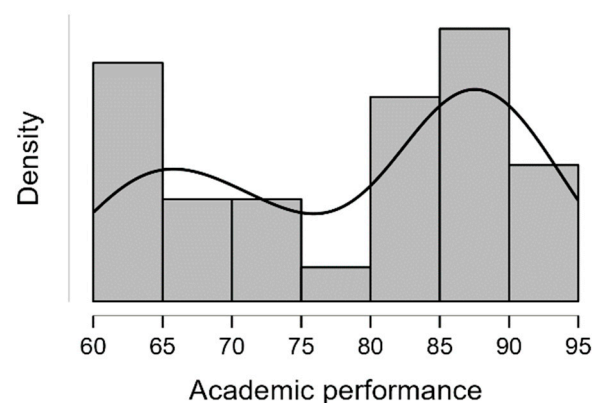


Figure 1. The Distribution of academic performance of the experimental group.

Therefore, the Mann–Whitney U test was used for statistical analysis of academic performance. As shown in Table 8, the results of the Mann–Whitney U test for the exam assessing learners' academic performance ($W = 675$, $p = 0.014$, rank-biserial correlation = 0.361) indicated a significant difference in academic performance between the experimental group and the control group, with the effect size being at a medium level, which means LLM-based feedback significantly affected learners' academic performance.

Table 8. Mann–Whitney U test of academic performance of two groups.

	W	<i>p</i>	Rank-Biserial Correlation	SE Rank-Biserial Correlation	95% CI for Rank-Biserial Correlation	
					Lower	Upper
Academic performance	675.000	0.014	0.361	0.146	0.090	0.582

For the Mann–Whitney test, the effect size is given by the rank-biserial correlation.

5. Discussion

5.1. The Implementation of LLM-Based Feedback in SRL

In this study, one of the LLMs, GPT 3.5, was integrated with the online platform to support learners learning Python. There were four types of general feedback provided by the GPT API: task-level feedback, process-level feedback, self-regulation feedback, and self-level feedback, which correspond to the four dimensions of SRL (cognition, behavior, metacognition, and motivation). The details of LLM-based feedback in the Python course of this study included the following aspects:

- Identification and explanation. LLM-based feedback can identify the error locations, provide error types, and explain the workings of programming concepts, algorithms, or code segments to help learners have a better understanding;
- Code improvement suggestion. To make the code correct, optimized, or clear, LLM-based feedback can offer improvement suggestions, such as code refactoring or alternative implementation methods;
- Self-regulation hint. LLM-based feedback can provide learners with hints and guidance for problem-solving, but avoid directly providing answers.
- Constructive evaluation. LLM-based feedback can evaluate learners objectively, and encouragingly, and help learners to reflect on learning.

To facilitate access to LLM-based feedback, prompts were designed based on these four types of feedback. This design and application of prompts have prompted reflection on related research and practice. While most prompt design studies focus on creating prompts that elicit complete and accurate responses from LLMs, in education, one-step prompts that provide easy answers may undermine the learning process and reduce the quality of learning. Therefore, exploring interactive and personalized prompt designs that focus on guiding the SRL process is essential for the effective application of LLMs in education.

5.2. The Effect of LLM-Based Feedback on SRL

About the second research question, we investigated the effects of LLM-based feedback on learners' SRL by examining differences between the two groups. The findings based on statistical analyses of learners' average assignment scores showed better cognitive and behavioral performance during SRL for learners who received LLM-based feedback in comparison to learners with no LLM-based feedback. However, the statistical effect size was only at a small level. Furthermore, the results indicated that learners who received LLM-based feedback performed better in motivation and metacognition (SRL strategy) than learners who did not have LLM-based feedback. The statistical effect sizes were at a medium level.

As we mentioned above, for the types and characteristics of errors and exceptions in Python, the online platform can give automatic feedback on syntax errors, and runtime errors, but it is not very effective in handling logic errors and various exceptions. However, the identification and resolution of such complex logical errors and exceptions is precisely the advantage of LLM-based feedback, which means LLMs have strong capabilities to provide learners with task-level feedback and process-level feedback, promoting learners'

cognition and behavior in SRL. Also, the prompt design facilitated interactive feedback between GPT API and learners in the experimental group, which makes GPT API more effective in supporting SRL, which likely contributed to the higher average assignment scores observed in the experimental group.

Moreover, learners in the experimental group exhibited greater motivation and self-regulation, suggesting that LLM-based feedback was well-received and considered beneficial. Interestingly, the mean motivation score was higher than the self-regulation score, indicating that while LLM-based feedback effectively enhanced motivation, guiding learners in self-regulation remains more complex, even with LLM support. This finding suggests that self-regulation feedback and prompts need further refinement to maximize their effectiveness.

5.3. The Effect of LLM-Based Feedback on Academic Performance

Concerning the third research question, we analyzed differences in academic performance as measured by learners' exam scores to explore the potential effects of LLM-based feedback. Our findings suggest a group difference in academic performance, indicating that LLM-based feedback may support academic performance by contributing to learners' SRL. This finding is consistent with the SRL model proposed by previous researchers [7]. The model suggests that SRL serves as a mediator through which feedback influences academic performance. Learners use externally provided feedback to self-regulate, including fostering motivational beliefs, setting learning goals, selecting appropriate learning strategies to achieve these goals, and ultimately enhancing academic performance.

Throughout the SRL process, learners require feedback to adjust and refine their learning approaches, thereby boosting academic performance. SRL emphasizes developing learners' abilities to adapt learning plans and strategies, enabling them to master learning methods and, ultimately, learn how to learn. Consistent with previous research, this study found that the effect of SRL on academic performance was at a medium level, supporting the need for further exploration of LLM-based feedback in education.

SRL is a critical skill for fostering lifelong learning, an essential component of sustainable development. Our findings suggest that LLM-based feedback can effectively support the development of SRL, thereby enabling learners to adapt and thrive in a rapidly changing world. The use of LLMs to deliver continuous and tailored feedback can lead to more efficient learning practices, reducing the reliance on human-led interventions without compromising quality. This aligns with the goal of promoting sustainable and scalable educational goals.

On the other hand, an unexpected result was found in this study. When LLM-based feedback was not available, some learners in the experimental group were able to complete the exam better with higher scores, while others were not (please see the details in Figure 1). Since the prompts for the early tasks not only helped to correct learners' misunderstandings but also provided additional information, this may be because LLM-based feedback played a key role in enhancing learners' learning process, apparently further facilitating their understanding. This knowledge acquisition in turn may have accelerated their problem-solving skills. However, when LLM-based feedback was unavailable, some other learners in the experimental group scored lower on the exam. The findings suggest that while LLM-based feedback may enhance learning and understanding, it might also lead to overreliance on LLMs, hindering performance when such feedback is not available. If this is the case, further investigation is warranted in future studies.

6. Conclusions

This study utilized the GPT-3.5 model to provide feedback to learners in a programming course, focusing on enhancing their self-regulated learning (SRL) to support learners'

sustainable development. The results indicated that LLM-based feedback can significantly improve learners' SRL in terms of motivation, learning strategies, and cognitive and behavioral aspects, which, in turn, positively impacts their academic performance, and therefore improves learners' sustainable development.

Providing effective feedback for programming learning is challenging due to the variety of errors, exceptions, and complex issues that arise in different programming tasks. Leveraging the natural language capabilities of LLMs, such as the GPT model, offers a promising solution to these challenges. The findings of this study suggest that integrating LLM-generated feedback into computer programming education can positively influence learners' SRL, ultimately leading to improved learning outcomes. While this study utilized the GPT-3.5 model, fine-tuning the model could further enhance the quality of generated prompts, thereby supporting more effective programming learning on a broader scale.

However, this study has some limitations. Firstly, although 10 weeks was sufficient in terms of intensity, learners were introduced to LLM-based feedback tools for the first time, so the 10-week period may be relatively short compared to the time they have spent using traditional tools. The lack of familiarity with these tools may have contributed to the limited effectiveness observed. Secondly, as this was the researchers' first attempt to integrate LLMs into SRL, there were challenges in both the integration approach and prompt design. Future research should investigate learners' experiences and reflections on using LLM-based feedback to refine these tools for better educational outcomes. Finally, the study was limited to a Python programming course for computer science majors, restricting the generalizability of the findings. While the results of this study are promising, they may not directly generalize to other educational settings, disciplines, or different LLMs, especially those with distinct characteristics or applications. Therefore, future research should explore how these findings can be applied to different educational contexts, including disciplines outside of computer science, and with other types of LLMs.

To address these limitations, future studies should extend the experimental period and refine the integration of LLMs to ensure more accurate results. Additionally, expanding the application of these feedback tools to other disciplines and testing their effectiveness with different LLMs will be crucial for understanding their broader impact. This is a worthwhile area for further exploration.

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Data Availability Statement: The data presented in this study are available on request from the corresponding author. The data are not publicly available due to anonymity requirement.

Conflicts of Interest: The authors declare no conflicts of interest.

Appendix A

I. Motivation items based on MSLQ:

1. On the programming platform, I prefer challenging content because it allows me to learn new things (based on MSLQ-No. 1).
2. If I study in a proper way, I will be able to master the content of this course (based on MSLQ-No. 2).
3. On the programming platform, I like learning content that arouses my curiosity, even if it is difficult to learn (based on MSLQ-No. 16).
4. I am confident that I can successfully complete the homework and exams on the programming platform (based on MSLQ-No. 21).
5. For me, the most satisfying thing about the programming platform is understanding the content as deeply as possible (based on MSLQ-No. 22).
6. Taking into account the difficulty of the content, the teacher and my ability, I think I will do well in this course (based on MSLQ-No. 31).

II. Learning strategies (metacognition) items based on MSLQ:

1. I often question the content offered to me in the programming platform to determine if I am convinced (based on MSLQ-No. 38).
2. When using the programming platform to learn, I will actively communicate with the programming platform and make marks to try to find the most important ideas (based on MSLQ-No. 42).
3. When learning using the programming platform, I integrate information from different sources, such as lectures, textbooks, and discussions (based on MSLQ-No. 53).
4. When learning with the programming platform, I try to think deeply about a topic and decide what I should learn from it, rather than just reading it once (based on MSLQ-No. 61).
5. When learning using the programming platform, I try to relate it to what I already know (based on MSLQ-No. 64).
6. When I cannot understand what I am learning, I turn to the programming platform for help (based on MSLQ-No. 68).
7. Whenever I use the programming platform to solve a problem, I interact with the platform about possible alternatives (based on MSLQ-No. 71).
8. When learning using the programming platform, I set goals for myself based on the platform's feedback to guide the activities of each learning phase (based on MSLQ-No. 78).

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